

The spring science evidence exchange

Teacher guide and worked answers

Use evidence to explain life cycles and apply selected spring science ideas in a teacher-authored checkpoint.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | C39

Consolidate life cycles; evidence spring science (Entry Level Science / UAS).

Intentional consolidation, teacher-authored evidence checkpoint; no UAS or qualification completion claim. Original secondary-age exemplar at an upper-KS2 conceptual level; no qualification approval or completion claim.

Prior learning

Recall life-cycle stages, heart pumping and material testing.

Use values with units to support a claim.

Success evidence

Describe a life-cycle sequence and a meaningful comparison.

Explain a body-system job accurately.

Use material-test data and state a limit.

Equipment

Training sheet, one checkpoint route, pencils; optional calculator.

Preparation

Give only shared training page 1 during I do and We do. Keep route pages 2-4 face down until You do.

Do not rehearse the checkpoint's animal names, numerical values or answers before independent work.

Read or scribe neutrally; record any adult prompts separately.

Practical care

No animal handling, exercise measurements, heat or electrical practical is needed.

Use private feedback and flexible support; do not infer qualification completion from this lesson.

Ready to teach alternative

All checkpoint evidence is supplied on the selected paper route. No internet or practical materials are needed.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Show the science you can explain

Slide	Stage	Minutes	Focus
2	Retrieval	2-5	Remember the evidence habits
3	I do	5-13	I model how to answer, not the checkpoint
4	I do	Within stage	I inspect the comparison conditions
5	I do	Within stage	I bound the training claim
6	We do	13-21	Training editor: improve the response
7	We do	Within stage	Choose useful feedback
8	We do	Within stage	Prepare for independent work
9	Hinge	21-24	Which is a useful evidence sentence?
10	You do	24-34	Your independent investigation
11	You do	Within stage	Use the evidence
12	You do	Within stage	Check before sharing
13	Review	34-38	Compare and improve
14	Review	Within stage	A question worth investigating
15	Exit	38-40	Show the science you can explain

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Show the science you can explain

Explain that this is feedback on selected taught content, not an official exam or grade predictor. Keep selected route pages face down.

2 Retrieval Remember the evidence habits

Units give meaning; bounded context limits a claim; yes, neutral reading supports access. Do not review the checkpoint's exact factual answers.

3 I do I model how to answer, not the checkpoint

Think aloud using the training car example only. The purpose is a response routine; the independent lifecycle, body and materials cases are not shown or solved here.

4 I do I inspect the comparison conditions

Shared I do time. Trace the two ramps. This is a schematic, not a scale drawing. Model the control question: What else could change the result? Do not introduce checkpoint values.

5 I do I bound the training claim

Shared I do time. Model: In these constructed trials, the car travelled farther on surface A, for example 60 cm versus 40 cm in trial 1. Repeat conditions matter. Avoid using the independent insulation data.

6 We do Training editor: improve the response

Use the toy-car dataset only. A exceeds B in both trials; 60/40 or 58/42 cm. Replace always with in these sample trials and suggest more repeats or conditions.

7 We do Choose useful feedback

Shared We do time. Add a numerical comparison that answers the question; speed is not the science goal. Allow private written choice.

8 We do Prepare for independent work

Shared We do time. Hand out or reveal pages 2, 3 or 4 only after the hinge. No answer key or teacher page should be visible.

9 Hinge Which is a useful evidence sentence?

A unsupported preference. B accurate values/context. C universal overclaim. This tests evidence language, not checkpoint science answers.

A. A is better because it looks nicer. - Appearance is not evidence about the measured distance.

B. In trial 1, the car travelled 60 cm on A and 40 cm on B. - It gives relevant values, units and a clear boundary.

C. The car must always travel farther on A in every situation. - Two training trials do not establish every situation.

10 You do Your independent investigation

10 minutes across the three You do slides. Supply shared page 1 and one selected route page. All routes target the stated science objective; a labelled drawing or independent dictated explanation is valid.

11 You do Use the evidence

Shared You do time. Ask pupils which evidence supports their explanation. Read prompts or scribe without supplying the scientific conclusion.

12 You do Check before sharing

Shared You do time. Use the route-specific worked answers to identify one precise next step.

13 Review Compare and improve

Collect the independent checkpoint before discussing its answers. Use the separate teacher answers to give private criterion-based feedback. If pupils annotate afterwards, keep the original response visible so the first attempt is distinguishable.

14 Review A question worth investigating

Lundy cycle: invite written, spoken or pointed suggestions. Select one feasible question and explain how the pupil contribution changes the next example or investigation.

15 Exit Show the science you can explain

Exit is reflection after submission, not a second independent score. Accept a corrected lifecycle, body or material explanation with relevant evidence.

Answers and assessment

R1-R3 / training

Units give a number meaning; in this test bounds a claim; neutral reading can support access without supplying a conclusion. W1 A. W2 60 vs 40 or 58 vs 42 cm. W3 in these sample trials, the car travelled farther on A; repeat and test further conditions. Useful feedback asks for evidence, not writing speed.

Hinge A-C

A preference is not distance evidence. B correct. C overclaims two trials.

S1-S5

S1 ladybird. S2 both eggs/adults; ladybird has pupa, frog does not. S3 pumps blood. S4 thick sleeve. S5 28 °C remains closer to the 30 °C start than 24 °C; it slowed cooling, not stopped all energy transfer forever.

N1-N4

N1 both eggs and reproductive adults; ladybird has pupa whereas dragonfly has nymph and no pupa. Adult → eggs represents new offspring, not the same individual becoming an egg. N2 digestion produces small absorbable nutrients which blood transports; heart pumps blood. N3 falls 6 °C and 2 °C; fabric slowed energy transfer out to the cooler room. N4 one trial limits confidence; repeat matched conditions. Other relevant limitations and specific improvements accepted.

T1-T4

T1 dragonfly nymph becomes adult without pupa; platypus is an egg-laying mammal. T2 digestive organs break some food molecules down; small nutrients are absorbed and carried in blood; heart pumps. T3 6 °C versus 2 °C; sleeved bottle still cools, so transfer continues. T4 same fabric at two measured thicknesses, matching bottles, fill, starting temperature and room; record temperature after a fixed duration and repeat.

Criteria and diagnostic response

Record each area as secure, developing or revisit, without an official grade. Life cycles secure: a valid matched comparison and generational meaning; body systems secure: correct pumping and absorption/transport link; materials secure: correct paired values, slows transfer and bounded conclusion. A correct choice without explanation is developing evidence. Re-teach only the area indicated: stage cards, two-system substance route, or warmer-to-cooler arrow.

E1-E4 / reflection

Accept a reasoned self-assessment tied to the submitted work. Corrections must be scientifically accurate and labelled as after feedback. A useful next task names a concept, such as distinguishing nymph from pupa or drawing energy direction, rather than simply “do more science”.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

This is original formative assessment of selected Spring content, not an official awarding-body paper or proof of qualification completion.

The training car example develops the evidence-response routine. The checkpoint uses different organisms, body-system questions and cooling values; do not reveal or rehearse these beforehand.

Interactive training uses only page 1. Collect checkpoint work before answer feedback; preserve the first attempt when pupils correct afterwards.

Access and responsive teaching

Supported

Short choices and sentence frames; same three science areas.

Standard

Explain three evidence cases with a comparison and numerical support.

Stretch

Evaluate overclaims and suggest a precise next check.

Regulation

Offer private writing, pointing, drawing or adult scribing. Allow pass-and-return for public questions and desk-based participation. Choose support from current learning evidence, not fixed ability labels.

Misconceptions to check

A correct choice alone always proves understanding.

An explanation or evidence link shows more than a guess.

Check: Ask which feature or number supports the choice.

A classroom checkpoint gives an official qualification result.

It samples taught ideas for feedback; qualification requirements need separate verification.

Check: Ask what the checkpoint was designed to show.

One material result proves what always happens.

A conclusion should name the tested conditions and relevant limits.

Check: Ask which extra test would strengthen the claim.

Word	Meaning
evidence	Information used to support or challenge a claim.
explanation	An account of how or why something happens.
comparison	A similarity or difference using the same feature.
criterion	A feature used to judge whether work meets the goal.

Scientific references

[Monarch Joint Venture: Life cycle](#)

Checked September 2026. Named monarch sequence: egg, larva, pupa, adult; not a rule for all insects.

[Arizona State University: Incomplete metamorphosis](#)

Checked September 2026. Egg-nymph-adult is different from complete metamorphosis; dragonfly nymphs are aquatic. The checkpoint supplies the necessary species clues.

[NHLBI: The heart](#)

Heart pumping and blood transport reference, checked September 2026.

Royal Society of Chemistry: Separation techniques

Checked September 2026. Filtration, evaporation and separation misconceptions. Classroom activities are original and include a complete paper route.

US National Park Service: Dragonfly larvae

Checked September 2026. Aquatic dragonfly juveniles; insects with this incomplete metamorphosis route do not have a pupa.

The Wildlife Trusts: Lovely ladybirds

Checked September 2026. Explicit ladybird egg, larva, pupa and adult sequence.